DIPENDRANATH MAHATO (he/him/his)

Department of Mathematics
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Professional Appointments

- **The University of Texas Rio Grande Valley** — September 2026 - Present
  School of Mathematical and Statistical Sciences — ***Lecturer I***

Education

- **Doctor of Philosophy** — August 2021 – August 2026
  Mathematics, *Advisor: Dr. Tài Huy Hà* — ***Tulane University, United States***
- **Master of Science** — July 2018 – July 2020
  Major in Mathematics, *Advisor: Dr. D. K. Basnet* — ***Tezpur University, India***
- **Bachelor of Science** — August 2013 – April 2017
  Honours in Mathematics — ***University of Calcutta, India***

Research Interests

Commutative Algebra ○ Algebraic Combinatorics ○ Algebraic Geometry

- Interpolation Problem in Projective Spaces
- Algebraic and Combinatorial Properties of Various Ideals

Awards & Honours

- **Phillips Distinguished Graduate Research Fellowship** — 2025-2026
  Department of Mathematics, Tulane University — *New Orleans, Louisiana*
- **2nd Place in Poster Session Contest** — July 2025
  Mathematical Congress of the Americas (MCA 2025) — *Miami, Florida*

Publications & Preprints

- **Rational Symbolic Power of Ideals** — (Preprint) arXiv:2511.02732
  with S. Dey, T. Ha, P. Mantero — *Status: Under Review*
- **Demailly's Conjecture for general and very general points** — (Preprint) arXiv:2409.08535
  with S. Bisui — *Status: Under Review*

○ Ongoing Projects

- **k-Configurations in $\mathbb{P}^N$** — with S. Bisui, T. T. Nguyen
- **v-number of some Ideals** — with M. Kumar
- **Points in Multi-Projective Spaces** — with E. Guardo
- **Interpolation and Containment of Fat Point Ideals** — with T. Ha

○ Dissertations

- **Symbolic Powers of Ideals: From Interpolation to Rational Extensions** — Submission
  Doctoral Thesis — *Defended: July 30, 2026*
- **Insight into Gaussian Sums and its Application** — Submission
  Master's Thesis — *Defended: June 17, 2020*

Teaching Experiences

- **Instructor of Records** — **Tulane University**
  - *MATH 1210: Calculus I* — *Summer 2025*
  - *MATH 1210: Calculus I* — *Fall 2024*
- **Teaching Assistant** — **Tulane University**
  - *MATH 1210: Calculus I* — *Fall 2021, Spring 2025, Fall 2025*
  - *MATH 1220: Calculus II* — *Spring 2022*
  - *MATH 1230: Stats for Scientists* — *Fall 2022, Spring 2024, Summer 2026*
  - *MATH 3050/6050: Real Analysis* — *Spring 2023*
  - *MATH 3090/6090: Linear Algebra* — *Fall 2023*
  - *CMPS/MATH 2170: Discrete Mathematics* — *Fall 2023*
  - *MATH 1100: Introduction to Probability and Statistics* — *Summer 2023, Summer 2024*
- **Mentor** — **Directed Reading Programs**
  - *Nikhil S. Kumar (NIT-Tiruchirappalli, India) - Exploring Commutative Algebra* — *TWOPLES- Fall 2025*
  - *Imdadul Hussain (Tezpur University, India) - An Introduction to symbolic powers* — *TWOPLES- Spring 2025*

Invited Talks & Poster Presentations

○ Invited & Seminar Talks

- **Can we interpolate over fat points in Projective Space?** — Graduate Colloquium
  Department of Mathematics, Tulane University — *April 21, 2026*
- **Interpolation for points in $\mathbb{P}^N, N \geq 2$** — Algebra Seminar
  Center of Mathematical Sciences and Applications (CMSA) at Harvard University — *April 16, 2026*
- **Interpolation in Projective Space: From Reduced Points to Fat Points** — Algebra Research Seminar
  University of Arkansas (Spring 2026) [Host: P. Mantero] — *April 03, 2026*
- **Rational Symbolic Power of Ideals** — Commutative Algebra Seminar at Purdue
  Purdue University (Spring 2026) [Host: I. Swanson] — *March 25, 2026*
- **Going Beyond the Integer Symbolic Powers of Ideals** — AMS Spring Western Meeting 2026
  Boise State University (Special Session) — *March 07-08, 2026*
- **Unfolding the Rational Symbolic Powers of Ideals** — Joint Mathematics Meetings 2026
  Walter E. Washington Convention Center (Special Session) — *January 04-07, 2026*
- **Introduction to Rational Symbolic Powers** — Graduate Colloquium
  Department of Mathematics, Tulane University — *November 11, 2025*
- **Unfolding the Rational Symbolic Powers of Ideals** — AMS Fall Central Meeting 2025
  Saint Louis University (Special Session) — *October 18, 2025*
- **Interpolation Problem and Recent Studies on Demailly's Conjecture** — AMS Spring Western Meeting 2025
  California Polytechnic State Univeristy (Special Session) — *May 03, 2025*

- **Macaulay2 Workshop (Lightning Talk)** — Macaulay2 at Tulane 2025
  Department of Mathematics, Tulane University — *April 18, 2025*
- **Interpolation in Projective Spaces (Lightning Talk)** — Commutative Algebra in The South (CATS) 2025
  Department of Mathematics, Georgia State University — *April 06, 2025*
- **Interpolation of Projective Points under Algebraic Lens** — Math For All Virtual 2025
  Virtual — *April 05, 2025*
- **Interpolation Problem and Recent Studies on Demailly's Conjecture** — AMS Spring Central Meeting 2025
  University of Kansas (Special Session) — *March 29, 2025*
- **Developments in Interpolation Problem for Projective Spaces** — Algebra & Combinatorics Seminar
  Department of Mathematics, Tulane University — *March 14 (π-Day), 2025*
- **Interpolation Problem: Conjectures & New Developments** — CMSA-2025
  Department of Mathematical Sciences, Tezpur University — *January 03, 2025*
- **Recent Developments in Interpolation Problems & Demailly's Conjecture** — Graduate Colloquium
  Department of Mathematics, Tulane University — *November 05, 2024*
- **Overview of Generalized Interpolation Problem & its Algebraic Components** — TU Alumni Talk
  Department of Mathematical Sciences, Tezpur University — *May 24, 2024*
- **Generalized Interpolation Problems and Chudnovsky's Conjecture** — Graduate Colloquium
  Department of Mathematics, Tulane University — *April 25, 2023*
- **Algebraic Gaussian Sums and its Applications** — Graduate Colloquium
  Department of Mathematics, Tulane University — *October 04, 2022*

○ Poster Presentations

- **Introduction to Rational Symbolic Powers**
  - *CIMPA School PASCA 2.0*, *Centro de Investigación en Matemáticas (CIMAT)* — *May 18-29, 2026*
  - *CAAC 2026*, *Dalhousie University* — *January 23-25, 2026*
  - *SLAM 2026*, *University of Texas at Arlington* — *February 28-March 01, 2026*
- **Interpolation of Projective Points & Demailly's Conjecture**
  - *MCA 2025*, *Miami [* **2nd *Best Presentation*** *]* — *July 21-25, 2025*
  - *TRICS 2025*, *Tulane University* — *April 08-10, 2025*
  - *CATS 2025*, *Georgia State University* — *April 05-06, 2025*
  - *SLAM 2025*, *Arizona State University* — *March 01-02, 2025*
  - *KUMUNU 2024*, *University of Missouri* — *September 28-29, 2024*

Seminars/Conferences (Co-)Organized

- **American Mathematical Society: Sectional Meeting** — **Special Session on Commutative Algebra**
  2025 Fall Eastern Virtual Sectional Meeting — *Co-organizer: Souvik Dey*
- **American Mathematical Society: Sectional Meeting** — **Special Session on Commutative Algebra**
  2025 Fall Southeastern Sectional Meeting at Tulane University — *Co-organizers: Tài Huy Hà and Kuei-Nuan Lin*
- **Tulane University Research Talks (Mathematics)** — **AMS/AWM Faculty Talks**
  Fall 2023 & Spring 2024 — *Co-organizer: Rebecca Kahler*

Research & Travel Support

- **Research Support**
  - *Phillips Distinguished Graduate Research Fellowship, Tulane University* — *Spring 2026*
  - *Office of International Students & Scholars Scholarship, Tulane University* — *Spring 2025*
  - *Summer Research Support from Department of Mathematics, Tulane University* — *Summer 2022-2026*
- **Travel Supports**
  - *AMS Graduate Student Travel Grant for Central Sectional Meeting at Saint Louis University* — *Fall 2025*
  - *AMS Travel Grant for Mathematical Congress of the Americas (MCA) 2025* — *Summer 2025*
  - *SSE Dean's Travel Fund Award, Tulane University* — *Spring 2025*
  - *Department of Mathematics Travel Award, Tulane University* — *Fall 2024, Spring 2025, Spring 2026*
  - *GSSA Travel Support, Tulane University* — *Fall 2023, Spring 2025, Summer 2025*
  - *Host University/NSF Funded Travel Supports* — *Spring 2022 onwards*
    - Southern Regional Algebra Conference (SRAC) 2022
    - Texas Algebraic Geometry Symposium (TAGS) 2022
    - Southwest Local Algebra Meeting (SLAM) 2023-2026
    - MSRI/SLMath CMND Summer School 2023
    - Summer CAMP 2023, 2025
    - BrianFest 2023
    - UweFest 2024
    - KUMUNU 2024
    - Symmetries and Singularities in Texas 2025
    - Séminaire de Mathématiques Supérieures (SMS)
    - Combinatorial Algebra Meets Algebraic Combinatorics (CAAC) 2026
    - Purdue University
    - University of Arkansas
    - CMSA, Harvard University

Programming/Coding Skills

- Macaulay2/Singular
- LaTeX
- HTML & CSS
- C/C++

Outreach & Community Involvement

- **Academic Learning & Tutoring Center (ALTC), Tulane University**
  Math Department Teaching Assistant — *Fall 2023 - May 2026*
  - Facilitating problem-solving in both one-on-one and small group settings.
  - Assisting for Calculus I-III, Statistics, Linear Algebra, Real Analysis
- **K-12 STEM Outreach Program**
  Math Section Leader: GiST (Girls in STEM at Tulane) — *October 2022, April 2023*
  Math Section Leader: BATS (Boys At Tulane in STEM) — *April 2024, September 2024*
  - Welcoming 5th to 7th grade students for hands-on experience in STEM Fields.
  - Conducted different Math Models to learn advanced topics via stimulating activities.
- **American Mathematical Society (AMS) Tulane Student Chapter**
  Treasurer — *May 2022 – May 2023*
  President — *June 2023 – June 2024*
  - Organizer of AMS/AWM Faculty Talks, Tulane University.
  - Organized Annual Math BBQ, 2023 & 2024.
  - Compilation of Alumni Data in the Graduate Student Chapter Website.
- **South Asian Graduate Association (SAGA), Tulane University**
  Vice President — *June 2022 – May 2023*
  - Restarted the dormant association after CoVID-19.

- **Training and Placement Cell, Tezpur University**
  Departmental Coordinator — *August 2019 – July 2020*
  Secretary of Alumni Association — *July 2023 - Present*
  - Managing Academic & Job Allocation Database.
  - Organizer of 1st Math Alumni Meet, Tezpur University.

- **Students Science Council, Tezpur University**
  Departmental Representative — *October 2018 – December 2019*
  - Volunteered in Several Outreach Programs of $InSCIgnis$, 2019.

- **Department of Mathematical Science, Tezpur University**
  Departmental Coordinator — *October 2018 – December 2019*
  Volunteer — *January 2020 - July 2020*
  - Organized Math Events in $InSCIgnis$, 2019 and π-Day, 2019 & 2020 (Online).
  - Organized and Supervised Math Events and Quizzes in 2nd Assam Science Festival held at Tezpur.
  - Organized National Mathematics Day, the birthday of Srinivasa Ramanujan (Online).

Memberships

- American Mathematical Society (AMS) — 2021 onwards
- Society for Industrial and Applied Mathematics (SIAM) — 2021 onwards
- Mathematical Association of America (MAA) — 2022 onwards

Attended Conferences, Workshops & Seminars/Webinars (Selected)

- **CIMPA Summer School, 2026** — May 2026
  Centro de Investigación en Matemáticas (CIMAT) — *Guanajuato, Gto., Mexico*
- **Algebra Seminar** — April 2026
  Center of Mathematical Sciences and Applications (CMSA), Harvard University — *Cambridge, Massachusetts, United States*
- **Commutative Algebra Seminar at Purdue** — April 2026
  Purdue Univeristy — *West Lafayette, Indiana, United States*
- **Algebra Research Seminar** — April 2026
  Univeristy of Arkansas — *Fayetteville, Arkansas, United States*
- **AMS Sectional Meeting: 2026 Spring Western Sectional Meeting** — March 2026
  Boise State Univeristy — *Boise, Idaho, United States*
- **Southwest Local Algebra Meeting (SLAM) 2026** — February-March 2026
  University of Arlington — *Arlington, Texas, United States*
- **Combinatorial Algebra Meets Algebraic Combinatorics (CAAC) 2026** — January 2026
  Dalhousie University — *Halifax, Nova Scotia, Canada*
- **Joint Mathematics Meetings** — January 2026
  Walter E. Washington Convention Center — *Washington, D.C., United States*
- **AMS Sectional Meeting: 2025 Fall Central Sectional Meeting** — October 2025
  Saint Louis University — *St. Louis, Missouri, United States*
- **Mathematical Congress of the Americas (MCA) 2025** — July 2025
  InterContinental Miami, an IHG Hotel — *Miami, Florida, United States*
- **Séminaire de Mathématiques Supérieures (SMS) 2025** — June 2025
  Fields Institute, Univeristy of Toronto — *Toronto, Ontario, Canada*

- **AMS Sectional Meeting: 2025 Spring Western Sectional Meeting** — May 2025
  California Polytechnic State Univeristy — *San Luis Obispo, California, United States*
- **Macaulay2 Workshop** — April 2025
  Tulane University — *New Orleans, Louisiana, United States*
- **Tulane Research, Innovation and Creativity Summit (TRICS) 2025** — April 2025
  Tulane University — *New Orleans, Louisiana, United States*
- **Commutative Algebra in The South (CATS) 2025** — April 2025
  Georgia State University — *Atlanta, Georgia, United States*
- **Math For All in NOLA 2025 (Virtual Session)** — April 2025
  Tulane University — *[Hybrid] New Orleans, Louisiana, United States*
- **AMS Sectional Meeting: 2025 Spring Central Sectional Meeting** — March 2025
  University of Kansas — *Lawrence, Kansas, United States*
- **Southwest Local Algebra Meeting (SLAM) 2025** — February 2025
  Arizona State University — *Phoenix, Arizona, United States*
- **Symmetries and Singularities in Texas** — January 2025
  Texas A&M University — *College Station, Texas, USA*
- **Two-Day Conference on "Mathematical Sciences and Applications"** — January 2025
  Tezpur University — *[Hybrid] Tezpur, Assam, India*
- **KUMUNU** — August 2024
  University of Missouri — *Columbus, Missouri, USA*
- **UWE Fest 2024 (On the Occasion of Uwe Nagel's 60th Birthday)** — August 2024
  University of Notre Dame — *South Bend, Indiana, USA*
- **NCMW- Recent Trends in Commutative Algebra (2024)** — June 2024
  Indian Institute of Technology, Bombay — *Mumbai, India*
- **International Conference on Local Rings and Singularities** — June 2024
  Indian Institute of Technology, Bombay — *Mumbai, India*
- **Southwest Local Algebra Meeting (SLAM) 2024** — February 2024
  University of Oklahoma — *Norman, Oklahoma, United States*
- **BrianFest - Celebrate the Accomplishments of Prof. Brian Harbourne** — August 2023
  University of Nebraska-Lincoln — *Lincoln, Nebraska, USA*
- **MSRI/SLMath CMND Summer School** — May -June 2023
  University of Notre Dame — *South Bend, Indiana, USA*
- **Commutative Algebra in The South (CATS) 2023** — April 2023
  University of Alabama — *Tuscaloosa, Alabama, USA*
- **Southern Regional Algebra Conference (SRAC) 2023** — March 2023
  Tulane University — *New Orleans, Louisiana, USA*
- **Canadian Western Algebraic Geometry Symposium (CaWAGS)** — March 2023
  Simon Fraser University — *Burnaby, British Columbia, Canada*
- **Southwest Local Algebra Meeting (SLAM) 2023** — March 2023
  University of North Texas — *Denton, Texas, USA*
- **Southern Regional Algebra Conference (SRAC) 2022** — March 2022
  Georgia College & State University — *Milledgeville, Georgia, USA*
- **Combinatorial Algebra meets Algebraic Combinatorics (CAAC- 20th)** — January 2022
  The Fields Institute — *Virtual*
- **Combinatorial Algebra meets Algebraic Combinatorics (CAAC- 19th)** — January 2021
  Université du Québec à Montréal — *Virtual*
- **Advanced Instructional School on Commutative Algebra** — January 2021
  IIT Gandhinagar & IISC Bangalore with Advanced Training Schools — *Virtual*